1.3.7 Equivalence of matrix norms

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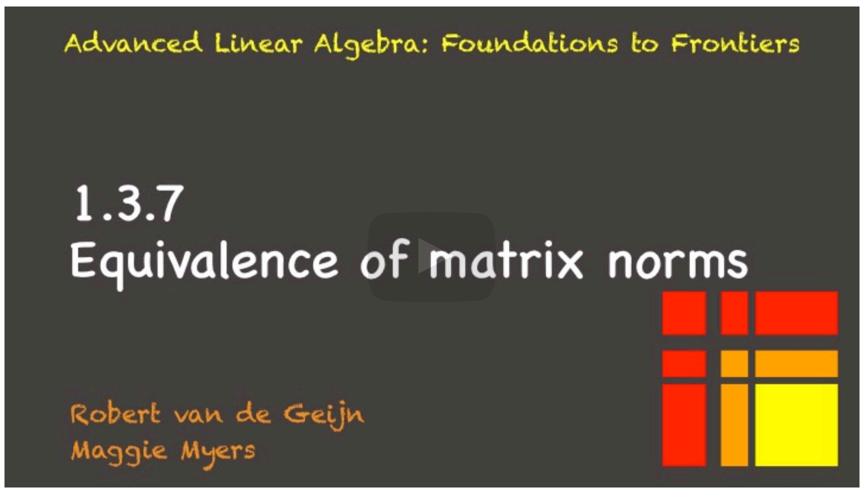
Week 01 due Jan 9, 2025 09:15 IST

Video



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Start of transcript. Skip to the end.

When we talked about vector norms, we made a big deal out of the fact that if

a vector was considered small in one norm, then it was also small in all other

norms. And if it was large in one norm, then it was large in all other norms. And

we called that equivalence

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Reading assignment

1.0/1.0 point (graded)

Read Unit 1.3.7 of the notes. [LINK]

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? why sqrt(n) vs sqrt(m)? 2

In the matrix of equivalences below, it says that norm2(A) <= sqrt(n)norm1(A), where n is the # ...

Homework 1.3.7.1.

1/1 point (graded)

Fill out the following table:

A	$\|A\|_1$	$\|A\|_\infty$	$\|A\|_F$	$\|A\|$
$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$	1 ✔ Answer: 1	1 ✔ Answer: 1	1.732 ✔ Answer: sqrt(3)	1 ✔
$\begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$	4 ✔ Answer: 4	3 ✔ Answer: 3	3.464 ✔ Answer: sqrt(12)	3.4 ✔ Ansv
$\begin{pmatrix} 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \end{pmatrix}$	3 ✔ Answer: 3	1 ✔ Answer: 1	1.732 ✔ Answer: sqrt(3)	1.7 ✔

For full solution see notes: [LINK]

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Homework 1.3.7.2

1/1 point (graded)
Given $A \in \mathbb{C}^{m \times n}$ show that $\|A\|_2 \leq \|A\|_F$.

Hmmm, actually, this is really easy to prove once we know about the SVD... Hard to prove without it. So, this problem will be moved...

☑ done

✔

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Homework 1.3.7.3

1/1 point (graded)
Let $A \in \mathbb{C}^{m \times n}$. The following table summarizes the equivalence of various matrix norm:

	$\|A\|_1 \leq \sqrt{m}\|A\|_2$	$\|A\|_1 \leq m\|A\|_\infty$	$\|A\|_1 \leq \sqrt{m}\|A\|$
$\|A\|_2 \leq \sqrt{n}\|A\|_1$		$\|A\|_2 \leq \sqrt{m}\|A\|_\infty$	$\|A\|_2 \leq \|A\|_F$
$\|A\|_\infty \leq n\|A\|_1$	$\|A\|_\infty \leq \sqrt{n}\|A\|_2$		$\|A\|_\infty \leq \sqrt{m}\|A$
$\|A\|_F \leq \sqrt{n}\|A\|_1$	$\|A\|_F \leq ?\|A\|_2$	$\|A\|_F \leq \sqrt{m}\|A\|_\infty$	

For each, prove the inequality, including that it is a tight inequality for some nonzero A.

(Skip $\|A\|_F \leq ?\|A\|_2$: we will revisit it in Week 2.)

☑ done

✔

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